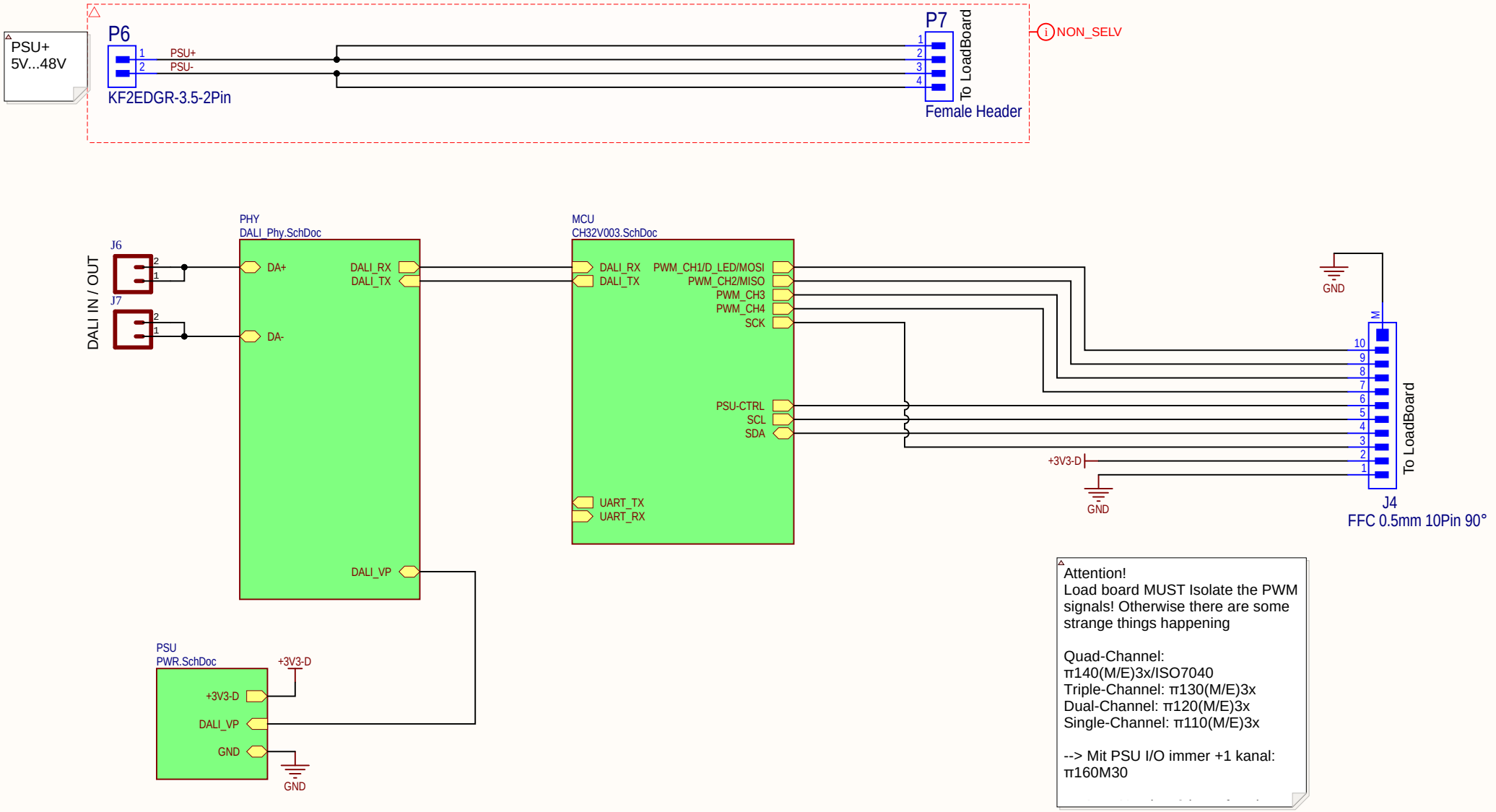
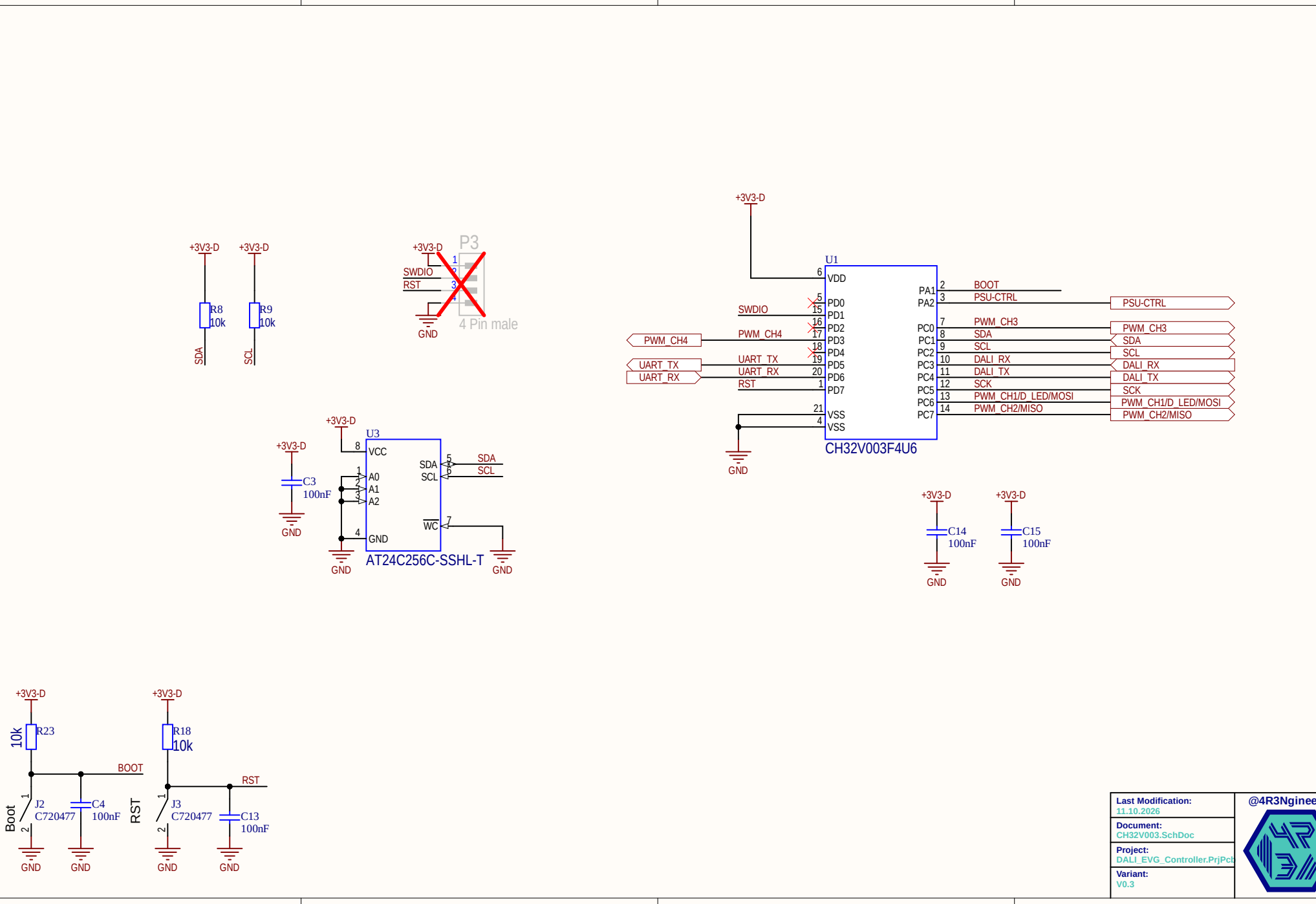
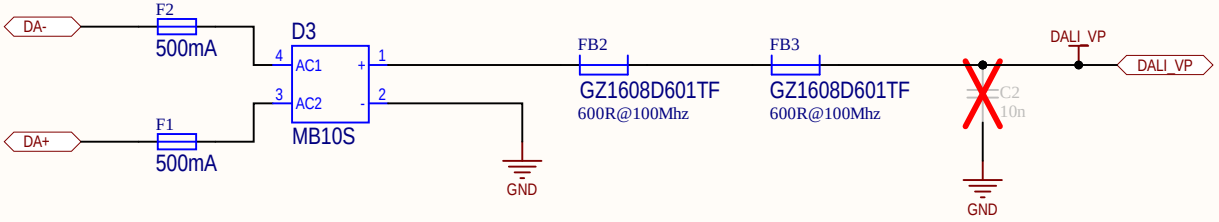

PSU+
5V...48V
P6
1 PSU+
2 PSU-
KF2EDGR-3.5-2Pin
P7
To LoadBoard
Female Header
NON_SELV
PHY
DALI_Phy.SchDoc
J6
J7
DALI IN / OUT
DA+
DA-
DALI_RX
DALI_TX
DALI_VP
MCU
CH32V003.SchDoc
DALI_RX
DALI_TX
PWM_CH1/D_LED/MOSI
PWM_CH2/MISO
PWM_CH3
PWM_CH4
SCK
PSU-CTRL
SCL
SDA
UART_TX
UART_RX
GND
+3V3-D
J4
To LoadBoard
FFC 0.5mm 10Pin 90°
PSU
PWR.SchDoc
+3V3-D
DALI_VP
GND
Attention!
Load board MUST Isolate the PWM signals! Otherwise there are some strange things happening
Quad-Channel:
π140(M/E)3x/ISO7040
Triple-Channel: π130(M/E)3x
Dual-Channel: π120(M/E)3x
Single-Channel: π110(M/E)3x
--> Mit PSU I/O immer +1 kanal:
π160M30
Last Modification:
11.10.2026
Document:
EVG_Controller.SchDoc
Project:
DALI_EVG_Controller.PrjPcb
Variant:
V0.3
@4R3Ngineering

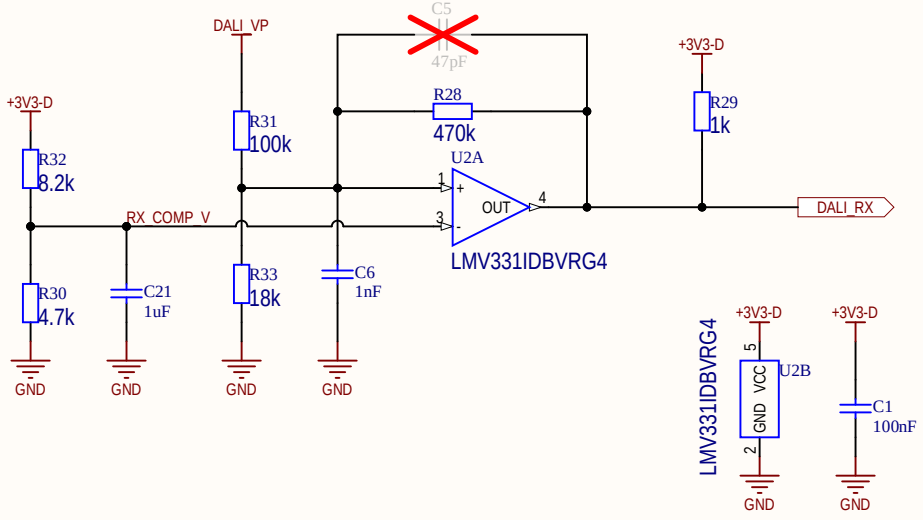

+3V3-D
R8
10k
SDA
+3V3-D
R9
10k
SCL
+3V3-D
P3
SWDIO
RST
GND
4 Pin male
+3V3-D
U1
VDD
PD0
PD1
PD2
PD3
PD4
PD5
PD6
PD7
VSS
VSS
SWDIO
PWM_CH4
UART_TX
UART_RX
RST
PA1
PA2
PC0
PC1
PC2
PC3
PC4
PC5
PC6
PC7
BOOT
PSU-CTRL
PWM_CH3
SDA
SCL
DALI_RX
DALI_TX
SCK
PWM_CH1/D_LED/MOSI
PWM_CH2/MISO
CH32V003F4U6
GND
+3V3-D
U3
VCC
A0
A1
A2
GND
SDA
SCL
WC
AT24C256C-SSHL-T
C3
100nF
+3V3-D
C14
100nF
+3V3-D
C15
100nF
+3V3-D
10k
R23
BOOT
Boot
J2
C720477
C4
100nF
+3V3-D
R18
10k
RST
RST
J3
C720477
C13
100nF
Last Modification:
11.10.2026
Document:
CH32V003.SchDoc
Project:
DALI_EVG_Controller.PrjPcb
Variant:
V0.3
@4R3Ngineering

C7...C10
Bypass-Caps für GND-LF

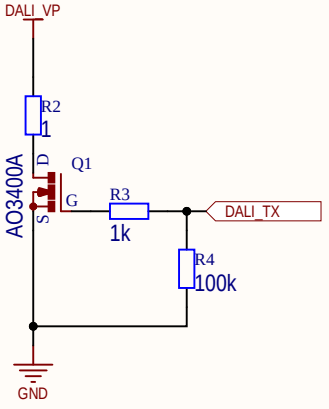
FB2/FB3/C2:
HF-Noise Filtering



DALI Receiver

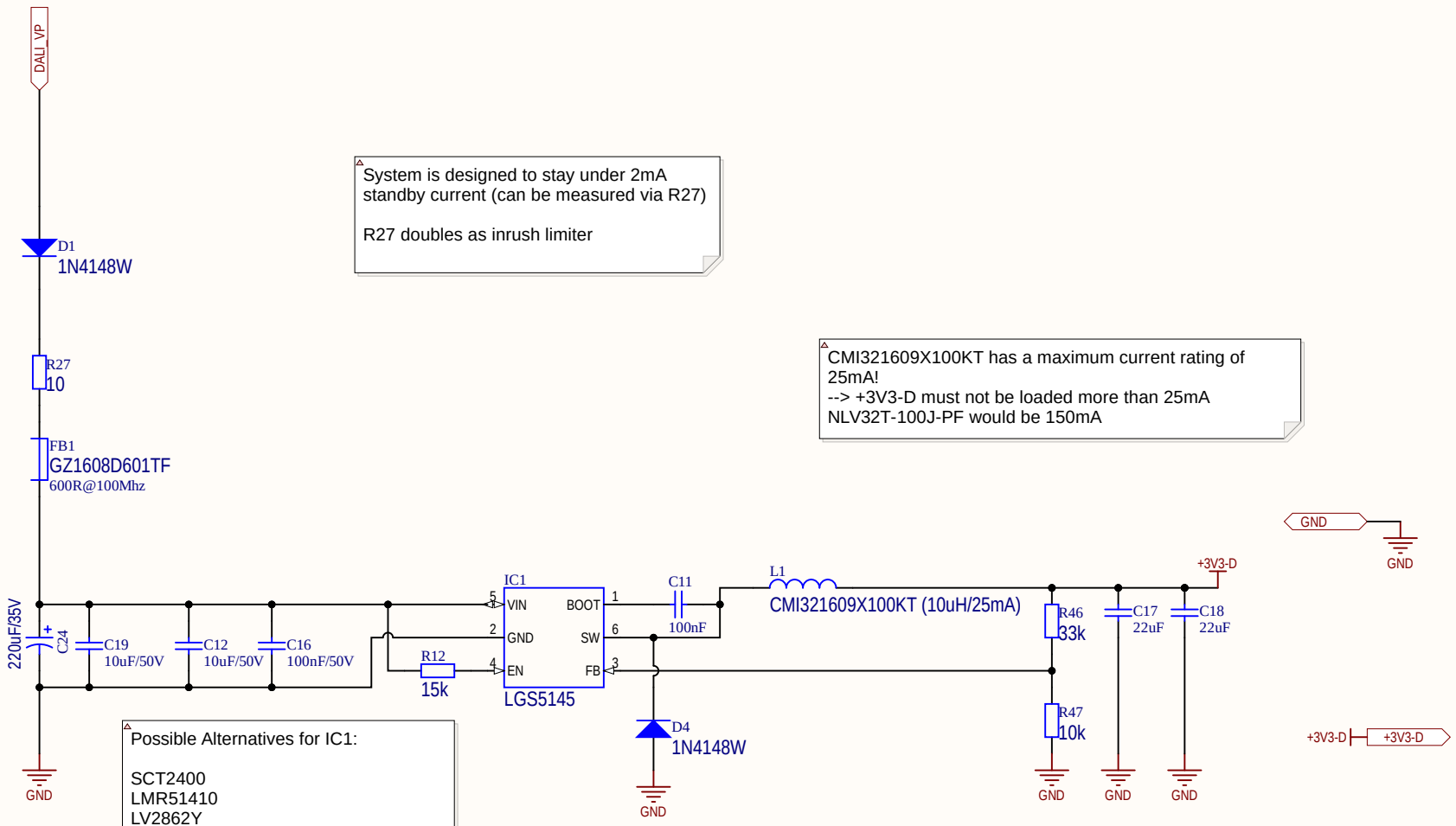


DALI Sender



Last Modification:	11.10.2026
Document:	DALI_Phy.SchDoc
Project:	DALI_EVG_Controller.PrjPcb
Variant:	V0.3

@4R3Ngineering



Last Modification:	11.10.2026
Document:	PWR.SchDoc
Project:	DALI_EVG_Controller.PrjPcb
Variant:	V0.3

@4R3Ngineering